



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Integrated Bachelor's-Master's Program
End-Semester Examination-Autumn 2025

Subject: Fundamentals of Molecular Cell Biology
Full Marks: 50

Subject Code(s): BIS 2101
Time Allotted: 3 h

Part-A

Answer any 25 questions:

1×25 = 25

1. What is another name for cell signaling?
2. Name any two types of cell signaling.
3. What type of signaling uses hormones that travel through the bloodstream?
4. Which type of signaling occurs between neighbouring cells?
5. What are the three main structural parts of a G protein-coupled receptor (GPCR)?
6. What enzyme converts ATP into cAMP?
7. Which enzyme breaks down cAMP into AMP?
8. Name any two examples of receptor tyrosine kinases (RTKs).
9. Which small GTP-binding protein is activated by RTKs?
10. Name any one signal protein that acts through the JAK-STAT pathway.
11. What are the two main categories of cell death?
12. Name the two main pathways of apoptosis.
13. Name any two death receptors involved in the extrinsic apoptotic pathway.
14. Which enzyme family mainly carries out apoptosis?
15. What type of cell death is associated with inflammation due to cell rupture?
16. Which protein is phosphorylated by RIPK3 during necroptosis?
17. Name the enzyme responsible for citrullination of histones during NETosis.
18. What is cancer?
19. Name any two major causes of cancer.
20. What are oncogenes?
21. What are tumor suppressor genes?
22. Name any two hallmarks of cancer proposed by Hanahan and Weinberg (2000).
23. What is the role of telomerase in cancer cells?
24. What is angiogenesis, and why is it important for tumor growth?
25. What does "metastasis" mean in cancer?
26. Name two examples of carcinogens that can cause cancer.
27. Which virus is associated with cervical cancer?
28. What is the Warburg effect?
29. Mention any one cancer type treated effectively by imatinib (Gleevec).
30. Name any two commonly used platinum-based chemotherapy drugs.

P.T.O.

Part-B

31. Define the following (any three):

1×3 = 3

- a) Transcription bubble
- b) Functional Proteome
- c) Zeitgebers
- d) siRNAs

32. Answer the following (any four):

3×4 = 12

a) Mention the major differences of DNA and RNA. What are the major structural characteristics of eukaryotic mRNA?

1½+1½ = 3

b) What is jet lag? Why effect of jetlag is more profound in adults than a newborn?

1+2 = 3

c) Name the tRNA binding sites present in an assembled ribosome. How elongation factors of prokaryotic cells ensure continuous aminoacyl tRNA delivery during translation.

1+2 = 3

d) How many types of RNA polymerases are found in eukaryotic cells. Mention their functions.

1½+1½ = 3

e) What is major histocompatibility complex? How do the different classes of this complex help in acquired immunity?

1+2=3

33. Explain the following (any three):

2×3 = 6

- a) Coding strand of DNA is not the template strand for transcription.
- b) 30S subunit of ribosome cannot bind to the 50S subunit before binding to the mRNA.
- c) Transcription and translation occur simultaneously in prokaryotic cells but not in eukaryotic cells.
- d) Flip-flop mechanism is involved in the sleep/waking transition.

34. Describe the following (any two):

2×2 = 4

- a) Initiation of translation in eukaryotic cells.
- b) Secondary structure of tRNA
- c) Transcription termination in prokaryotic cells.